Light

GCSE Computer Science - Data Representation: Media (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of a character set?

- ○ **A.** To assign a unique code to each character
- ○ **B.** To define fonts for printing
- ○ **C.** To compress text files
- ○ **D.** To encrypt messages

Q2 (1 mark)

What is the smallest individual unit of a digital image?

- ○ **A.** A byte

- ○ **B.** A frame
- ○ **C.** A bit
- ○ **D.** A pixel

Q3 (1 mark)

What does each sample in digital audio represent?

- ○ **A.** The amplitude of the sound wave at a moment in time
- ○ **B.** The total length of the recording
- ○ **C.** The file format being used
- ○ **D.** The number of audio channels

Q4 (1 mark)

What is a limitation of 7-bit ASCII?

- ○ **A.** It cannot represent numbers
- ○ **B.** It only works on modern computers
- ○ **C.** It uses too much storage space
- ○ **D.** It can only represent 128 different characters

Q5 (1 mark)

What does a colour depth of 8 bits mean?

- ○ **A.** The image is 8 pixels wide
- ○ **B.** Each pixel can be one of 256 colours
- ○ **C.** The image has 8 colours
- ○ **D.** The file size is 8 bytes

Q6 (1 mark)

CD quality audio uses a sampling rate of 44,100 Hz. What does this mean?

- ○ **A.** 44,100 samples are taken every second
- ○ **B.** The audio file is 44,100 bytes
- ○ **C.** The audio is 44,100 seconds long
- ○ **D.** There are 44,100 audio channels

Q7 (1 mark)

Why does Unicode include the same codes as ASCII for characters 0-127?

- ○ **A.** It was a mistake in the design
- ○ **B.** To make Unicode files smaller
- ○ **C.** Because there are no other characters to add
- ○ **D.** To ensure backwards compatibility with existing ASCII text

Q8 (1 mark)

In a bitmap image, what determines how much storage each pixel requires?

- ○ **A.** The file name
- ○ **B.** The image dimensions
- ○ **C.** The colour depth
- ○ **D.** The date the image was created

Q9 (1 mark)

What happens to sound quality if the sample resolution (bit depth) is reduced?

- ○ **A.** Quality decreases because amplitude is measured less precisely
- ○ **B.** Quality improves because files are smaller
- ○ **C.** Quality stays the same
- ○ **D.** The audio becomes louder

Q10 (1 mark)

In ASCII, consecutive letters have consecutive codes. If 'D' is 68, what is 'E'?

- ○ **A.** 67
- ○ **B.** 100
- ○ **C.** 70

- ○ **D.** 69

Q11 (1 mark)

What does a 1920 x 1080 image resolution mean?

- ○ **A.** The image is 1920 pixels wide and 1080 pixels tall
- ○ **B.** The file is 1920 kilobytes
- ○ **C.** The image uses 1920 colours
- ○ **D.** The image has 1080 bits per pixel

Q12 (1 mark)

Why can Unicode represent emoji but ASCII cannot?

- ○ **A.** ASCII is too old
- ○ **B.** ASCII does not work on phones
- ○ **C.** Emoji are not real characters
- ○ **D.** Unicode has many more character codes available

Q13 (2 marks)

Calculate the file size in bytes for a 50 x 40 pixel image with 16-bit colour depth.

- ○ **A.** 2,000 bytes

- B. 4,000 bytes
- C. 32,000 bytes
- D. 64,000 bytes

Q14 (2 marks)

An audio clip is recorded at 22,050 Hz with 16-bit resolution for 4 seconds. Calculate the file size in bits.

- A. 88,200 bits
- B. 352,800 bits
- C. 1,411,200 bits
- D. 2,822,400 bits

Q15 (2 marks)

In ASCII, uppercase 'A' is 65 and lowercase 'a' is 97. What is the code for lowercase 'e'?

- A. 69
- B. 99
- C. 101
- D. 133

Q16 (2 marks)

An image uses 5 bits per pixel. How many different colours can be displayed?

- ○ **A.** 5 colours
- ○ **B.** 10 colours
- ○ **C.** 25 colours
- ○ **D.** 32 colours

Q17 (2 marks)

Which audio settings produce better quality: 22,050 Hz with 8-bit, or 11,025 Hz with 16-bit?

- ○ **A.** 22,050 Hz with 8-bit because higher sampling rate is most important
- ○ **B.** 11,025 Hz with 16-bit because higher bit depth is most important
- ○ **C.** They are exactly the same quality
- ○ **D.** Cannot determine without knowing the file size

Q18 (2 marks)

A 1-bit image row has binary data 0101. If 1 = black and 0 = white, describe the pattern.

- ○ **A.** All black pixels
- ○ **B.** All white pixels
- ○ **C.** Alternating white and black pixels starting with white

- ○ **D.** Two white pixels followed by two black pixels

Q19 (2 marks)

A photo editor doubles both the width and height of an image. What happens to the file size?

- ○ **A.** File size doubles
- ○ **B.** File size triples
- ○ **C.** File size quadruples (becomes 4 times larger)
- ○ **D.** File size stays the same

Q20 (2 marks)

Why do Unicode text files often use more storage than ASCII text files with the same content?

- ○ **A.** Unicode files contain viruses
- ○ **B.** Unicode typically uses more bits per character than ASCII
- ○ **C.** ASCII files are automatically compressed
- ○ **D.** Unicode cannot store the same content as ASCII

Q21 (2 marks)

A stereo audio recording has two channels. How does this affect file size compared to mono?

- ○ **A.** File size is halved

- B. File size stays the same
- C. File size is doubled
- D. File size is tripled

Q22 (2 marks)

A smartphone camera is advertised as 48 megapixels. Approximately how many pixels is this?

- A. 48,000 pixels
- B. 480,000 pixels
- C. 4,800,000 pixels
- D. 48,000,000 pixels

Q23 (2 marks)

Using 7-bit ASCII, how many bits are needed to store the word 'CODE' (4 characters)?

- A. 4 bits
- B. 7 bits
- C. 28 bits
- D. 32 bits

Q24 (3 marks)

Calculate the file size in kilobytes (kB) for a 1600 x 1200 pixel image with 32-bit colour depth.

- ○ **A.** 768 kB
- ○ **B.** 6,144 kB
- ○ **C.** 7,680 kB
- ○ **D.** 61,440 kB

Q25 (3 marks)

Calculate the file size in megabytes (MB) for a 3-minute mono audio recording at 48,000 Hz with 24-bit resolution.

- ○ **A.** 25.92 MB
- ○ **B.** 17.28 MB
- ○ **C.** 8.64 MB
- ○ **D.** 51.84 MB

Q26 (3 marks)

Convert this 4x4 pixel 1-bit image to binary (1 = black, 0 = white): The image shows a vertical line down the left edge (leftmost column all black, rest white).

- ○ **A.** 1111 0000 0000 0000
- ○ **B.** 1000 1000 1000 1000
- ○ **C.** 0001 0001 0001 0001

- **D.** 1111 1111 1111 1111

Q27 (3 marks)

An image file is 480 kB. The image is 800 x 600 pixels. Calculate the colour depth.

- **A.** 4 bits
- **B.** 8 bits
- **C.** 16 bits
- **D.** 24 bits

Q28 (3 marks)

An audio engineer upgrades from 8-bit to 16-bit resolution while keeping the same 44,100 Hz sampling rate and 5-minute duration. By what factor does the file size increase?

- **A.** The file size stays the same
- **B.** The file size doubles
- **C.** The file size quadruples
- **D.** The file size increases by 8 times

Q29 (3 marks)

A programmer needs to store the text 'Hello' in a file. Compare using 7-bit ASCII versus a 16-bit Unicode format.

- ○ **A.** ASCII uses 35 bits; Unicode uses 80 bits; Unicode is about 2.3 times larger
- ○ **B.** Both use exactly 40 bits because they have the same characters
- ○ **C.** ASCII uses 40 bits; Unicode uses 40 bits; they are the same size
- ○ **D.** ASCII uses 35 bits; Unicode uses 70 bits; Unicode is exactly 2 times larger

Q30 (3 marks)

A security camera records images at 320 x 240 pixels with 8-bit colour. It saves one image per second. How much storage (in MB) is needed for 1 hour of images?

- ○ **A.** 27.6 MB
- ○ **B.** 276.5 MB
- ○ **C.** 2,765 MB
- ○ **D.** 27,648 MB

Total: **55** marks • Time: **30 min**

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